AI-Driven Diabetes Predictor System

BUSINESS ANALYSIS 3.2

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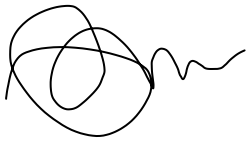
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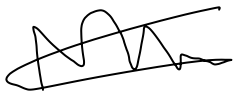
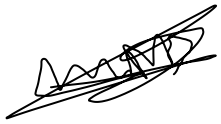
We, the undersigned, hereby declare that this project “AI Diabetes Detector" submitted is our own original work.

We confirm that:

- This project has been composed and written by us, and all sources of information, data, or code used have been properly acknowledged and cited.
- No part of this submission has been copied or plagiarized from any other group’s work.
- This work has not been previously submitted, in whole or in part, for any other module at the Vaal University of Technology or any other institution.

We understand that plagiarism is both unethical and illegal, and that it is punishable by the University leading to disciplinary action or expulsion.

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SECTION 1. Project Background & Theme Relevance

Developing and underdeveloped regions continue to face challenges in using Fourth Industrial Revolution (4IR) technologies to address everyday community needs. In line with the theme "An AI Solution for Industries," this project proposes a Python-based artificial intelligence solution for the healthcare industry. The system will use supervised machine learning to analyse health indicators and lifestyle information to predict whether a person is at risk of having diabetes or not. The system can help raise awareness and encourage people to seek appropriate healthcare. It is intended as a support tool and not as a replacement for professional medical assessment or diagnosis.

SECTION 2. Problem Definition

2.1 What is the Problem?

Diabetes is a long-term health condition that affects how the body regulates blood glucose. Some people may have diabetes without recognising the symptoms, and delayed testing or care can increase the chances of serious complications. Access to screening and health information may be limited for people who face financial, geographic, or healthcare-access barriers. Diabetes risk and status are associated with a range of factors, including body mass index, blood pressure, cholesterol, physical activity, general health, and other health and demographic indicators. Considering several factors together can be difficult to do consistently without analytical support. There is therefore an opportunity to develop an accessible machine-learning tool that analyses available health indicators and produces an estimated diabetes classification to support awareness and prompt appropriate follow-up. The prediction must not be treated as a confirmed medical diagnosis.

Source: Centres for Disease Control and Prevention (CDC)- Diabetes Basics

2.2 Local Municipality and Community Benefit

An AI-driven diabetes screening support tool could be useful to community health centres, local clinics, pharmacies, and health-awareness programmes. It could analyse a set of recorded health indicators and flag cases that may need further attention or formal testing by a qualified healthcare professional. Used responsibly, this could support health education and help direct attention toward people who may need medical attention.

The project will also demonstrate how data analysis and machine learning can be applied to public-health challenges. At the community level, the system will help communicate the importance of lifestyle and health indicators associated with diabetes. Any use with real people will require appropriate privacy and careful evaluation. Model predictions alone should not be used to make treatment decisions.

SECTION 3. System Aims and Business Objectives

3.1 Main Aim

The main aim of this project is to develop a Python-based machine learning application that classifies records into two outcome categories in the diabetes dataset: diabetes-negative (0) and diabetes-positive (1). The model will learn from the dataset's health and lifestyle inputs. The target column is Diabetes_binary, while the remaining 21 columns are testing input features.

3.2 Business Objectives and Success Criteria

- **Risk Classification:** The system should return a binary prediction corresponding to the dataset's Diabetes_binary target: 0 (diabetes-negative) or 1 (diabetes-positive). The 1 means the person is pre-diabetic or diabetic. The output must be described as a model estimate, not a medical diagnosis.
- **Efficiency:** Once trained, the model should return a prediction quickly enough for interactive preliminary screening. The team should measure prediction time during testing rather than assume that a particular response time has already been achieved.
- **Data Use and Accessibility:** The model relies only on the data provided. If we use this in the real world later, we will protect user privacy, ask easy-to-understand questions, and ensure users are guided to proper medical care.

3.3 Target Users

- **Main Users:** The project's intended users include healthcare-awareness programmes, pharmacies, and clinics.
- **Other Users:** Individuals who want to see if they may be at risk of having diabetes or not, maybe some related symptoms**.**

SECTION 4. Requirements, Constraints and Risks.

4.1 Functional and Non-Functional Requirements:

Functional:

- The system must accept the 21 health and lifestyle indicator inputs.
- Process them through the trained machine learning model and output a clear binary prediction (0 for diabetes-negative, 1 for diabetes-positive) plus a health recommendation if the result is 1.

Non-Functional:

- The application must return results quickly.
- Maintain data security standards when handling user information.

4.2 Project Constraints

- The project scope is strictly limited to supervised machine learning classification using the provided static dataset (253,680 records).
- Development and version control are managed via GitHub as required by the project guidelines, using Python libraries such as Scikit-Learn and Pandas.
- The system relies exclusively on self-reported survey features, meaning it cannot perform real-time blood glucose lab testing.

4.3 Ethical and Medical Risks

False Negatives:

- A critical risk is that the model might incorrectly classify a positive diabetes case as negative, potentially delaying necessary medical attention. To mitigate this, model recall will be closely monitored during evaluation.

Misinterpretation:

- Users might mistake the system's prediction for an official medical diagnosis. To address this, the clinics/pharmacies, etc., must explicitly state that the tool is strictly for awareness and preliminary screening, and users must be guided to consult qualified healthcare professionals.

SECTION 5. Initial Assessment of tools and techniques.

5.1 To build and test our AI diabetes detector, we are using the following tools and methods:

- **Programming Language and Libraries:** We are using Python because it is the industry standard for machine learning. We use Pandas and NumPy to clean and organise our data, and Scikit-Learn to train our machine learning models like Logistic Regression.
- **Version Control and Management:** All our group's code, files, and project tracking are managed on GitHub and the GitHub Project tool, as required by the project rules.
- **Testing and Evaluation:** To check how accurate and reliable our model is, we use testing metrics like accuracy, precision, recall, F1-score, and confusion

matrix, paying close attention to recall so we can minimise missed positive cases

SECTION 6. Machine Learning Specification & Architecture

6.1 Dataset Preparation and Feature Engineering

The supplied dataset contains 253,680 records and 22 columns. Diabetes_binary is the target variable. The remaining 21 columns are candidate model inputs. Before training, the team checked data types, missing or invalid values, class balance, duplicate records, and the dataset's coding scheme. The dataset's encoded values are interpreted using the relevant data dictionary rather than assuming that numeric codes are direct measurements.

The 21 input features in the supplied dataset are:

1. **HighBP: Indicates whether the patient reported high blood pressure.**
2. **HighChol: Indicates whether the patient reported high cholesterol.**
3. **CholCheck: Indicates whether the patient had a cholesterol check within the period defined by the dataset.**
4. **BMI: Body mass index, a measurement derived from height, weight and age.**
5. **Smoker: Indicates smoking status.**
6. **Stroke: Indicates whether the patient reported having had a stroke.**
7. **HeartDiseaseorAttack: Indicates a reported history of coronary heart disease or heart attack.**
8. **PhysActivity: Indicates whether the patient reported physical activity, as defined by the dataset.**
9. **Fruits: Indicates fruit consumption.**
10. **Veggies: Indicates vegetable consumption.**
11. **HvyAlcoholConsump: Indicates heavy alcohol consumption.**
12. **AnyHealthcare: Indicates whether the patient reported having healthcare coverage or access, as defined by the dataset.**
13. **NoDocbcCost: Indicates whether cost prevented the patient from seeing a doctor when needed.**
14. **GenHlth: Self-reported general health rating.**

15. MentHlth: Number of days of poor mental health reported within the dataset's reference period.

16. PhysHlth: Number of days of poor physical health reported within the dataset's reference period.

17. DiffWalk: Indicates difficulty walking or climbing stairs.

18. Sex: Indicates the gender of the patient, encoded numerically.

19. Age: Age category encoded numerically in the dataset, rather than a direct age in years.

20. Education: Education-level category encoded numerically.

21. Income: Income category encoded numerically.

Source: Kaggle dataset.

6.2 Supervised Learning Approach and Algorithms

The project will use supervised machine learning because the dataset contains a labelled outcome, Diabetes_binary. The models will learn patterns from the input features and predict one of two classes. The dataset should be split into training and test sets, with preprocessing fitted only on the training data to avoid data leakage. Model performance should be monitored using appropriate measures such as accuracy, precision, recall, F1-score, and a confusion matrix; recall is particularly relevant when considering the consequences of missed positive cases.

Primary Algorithm

- **Decision Tree:** A Decision Tree is suitable for binary classification because it uses a series of simple rules to predict a class. Its performance should be tested on data that was not used to train the model. The results should be interpreted carefully because patterns in the data do not prove that one factor causes another.

- **Comparison Algorithms**: We will be testing all the other supervised models to see which one is the most accurate when it comes to predicting the correct result. Its performance should be compared with Logistic Regression using the same data split and evaluation measures.

Source: TUTORIALS POINT AI with Python textbook.

SECTION 7. Theoretical Extensions

7.1 Time Series Analysis

If suitable repeated measurements become available in the future, the system can be extended to examine how health indicators change over time. Time Series Analysis can help describe changes in BMI, general health, or other recorded indicators across multiple visits. The current supplied dataset is not itself a sequence of repeated measurements for everyone, so time-series analysis would require additional data.

7.2 Solution Improvement Techniques

The model could be improved through careful validation, evaluation on data from different populations, and retraining only when suitable new data becomes available. Class imbalance, changing data patterns, and differences between the dataset and the intended users should be monitored. Any improvement should be confirmed through repeatable testing and documented performance measures.

7.3 Natural Language Processing (NLP) and Speech Recognition

A future interface could allow users to answer plain-language questions or speak their responses. Speech recognition and language support could make the system easier to use for people who have difficulty with text-based forms. Such an interface would need to explain questions clearly, protect personal information, and avoid implying that the result is a confirmed diagnosis.

7.4 Deep Learning Integration

If the project later has access to a larger, high-quality dataset, we could explore advanced AI methods (like neural networks). However, we should only use them if they prove to be better than simpler models and we can clearly explain how they work.

7.5 Conversational Chatbot / Softbot Interface

Our project will include a chatbot, through a web portal, which can guide users through the model's required inputs and allow users to ask questions where they might find confusion.

SECTION 8. REFFERENCES

Tutorials Point AI with Python textbook.

Google:

- **Source: Centres for Disease Control and Prevention (CDC)- Diabetes Basics**
- **Source: Centres for Disease Control and Prevention (CDC) — Diabetes risk factors**

Kaggle Diabetes binary dataset.